

Chapter # 6

Decision Trees and Decision Rules

Q. What is Classification?

⇒ Classification is a ^{form of} data analysis that extracts models describing important data classes.

⇒ Such models, called classifiers, predict categorical class labels.

Definition:

⇒ Classification involves learning a function that maps data items into predefined classes. ⇒ "Process of assigning a discrete label value to an unlabeled record."

Input:

⇒ Inductive learning algorithms receive a set of samples comprising attribute value vectors and corresponding classes.

Goal:

⇒ To create a classification model or classifier that predicts the class of a sample based on its ^{input} attributes.

Process

⇒ Samples are divided into predefined groups, assigning discrete label values to unlabeled records.

General Approach to Classification

How does Classification work?

⇒ Data classification is a two-step process:

1. Learning step
2. Classification step

1. Learning step

- where a classification model is constructed.
- Also known as Training Phase.
- Involves building a classifier using a training set made up of database tuples and their associated class labels.
- Classification algorithm builds a classifier.

Data representation

⇒ Each tuple is represented by an n -dimensional attribute vector, depicting measurements from database attributes.

Supervised learning

⇒ Because the class label of each training tuple is provided, this step is also known as supervised learning. Contrasting with unsupervised learning, class labels are unknown for training the classifier.

Class label attributes

⇒ The class label attribute is discrete-valued and unordered.

⇒ it is categorical in that each value serves as a category or class.

Purpose

⇒ Learning step aims to create a mapping or function that predicts the class label of a given tuple, typically represented as classification rules, decision trees or mathematical formulas.

2. Classification step:

- ⇒ Using the model to predict class labels for given data.
- ⇒ "What about classification accuracy?"

Evaluation of Classifier Accuracy:

- ⇒ The predictive accuracy of the classifier is estimated, usually using a test set independent of the training data.

Test sets

- ⇒ Consists of test tuples and their associated class labels, not used during the classifier construction.

Accuracy calculation:-

- ⇒ Percentage of test set tuples correctly classified by the classifier.

Application:

- ⇒ If the classifier's accuracy is acceptable, it can be used to classify future data tuples with unknown class labels.

Example: Classification rules learned from past loan applications can be applied to approve or reject new loan applicants.

Q. Define classifier?

- ⇒ Result of classification.
- ⇒ Classifier is a model that predicts one attribute - class of a sample.

Q. Graphical Interpretation of Classification?

Representations

- ⇒ A dataset with n -features is visualized as discrete points in an n -dimensional space.

Classification Rule:

- ⇒ Represented as hypercubes containing data points.
- ⇒ Multiple cubes can represent a single class, with conditions within a cube being AND-ed.

Cube size:

- ⇒ Indicates generality, with larger cubes covering more sample points.

Q. How Classification models Constructed?

Methodologies

⇒ Models can be obtained either through expert interviews or inductively from recorded classifications.

Expert-based Approach:

⇒ Relies on knowledge from relevant experts, despite known challenges.

Inductive Approach:

⇒ Constructs models by generalizing from specific examples, typical in data mining applications.

Q. Write different approaches for construction of a classification model?

Statistical Approach:

⇒ Summarizes statistical characteristics of sample sets.

Logical Approach:

⇒ Based on expressions that are evaluated as true or false by applying Boolean and comparative

Advantages of Logical models

⇒ Logical models offer accuracy and explanatory characteristics

⇒ Decision trees and decision rules are typical data-mining techniques that belong to a class of methodologies that give the output in the form of logical models.

Decision Trees

Overview:

- ⇒ Decision tree is a supervised learning method used in data mining for classification and regression methods.
- ⇒ Decision trees are efficient methods for producing classifiers from data.
- ⇒ They are widely used in logic-based methods for classification and regression.
- ⇒ Decision trees construct hierarchical models through recursive splits at decision nodes.

Structure:

- ⇒ A decision tree is like a flow-chart with:

Nodes: / Internal node / non-leaf node

- Representing tests on attributes.
- Decision trees have nodes where decisions are made.
- Univariate trees use only one attribute for testing at each internal node.
- Each decision is based on testing a certain attribute or characteristic.

Branches:

- Each branch represent an outcome of the test.
- Samples are partitioned along the branches.

Root Node:

- The topmost node in the tree, which represents the complete dataset.
- It is the starting point of the decision-making process.

Leaf/Terminal nodes:

- A node without any child nodes that indicates a class label or a numerical value.

⇒ Decision trees can deal with both categorical and numerical data.

Key factors

Entropy:

- Refers to a common way to measure impurity.
- it measures the randomness or impurity in data sets.
↓
uncertainty

formula:

$$\text{Entropy}(S) = -P(\text{Yes}) \log_2(\text{Yes}) - P(\text{No}) \log_2(\text{No})$$

Information Gains

- ⇒ Refers to the decline in entropy after the dataset is split.
- ⇒ Also called entropy reduction.
- ⇒ Building a decision tree is all about discovering attributes that returns the highest data gain.

formula:

$$I(n) = \text{Entropy}(S) - [\text{Weight Avg}] * \text{Entropy}(\text{each feature})$$

How are decision trees used for classification?

⇒ When a given tuple (a set of data), the attributes of the tuple are tested against the decision tree.

⇒ By tracing a path from the root node to a leaf node, the decision tree predicts the class label for that tuple.

⇒ DT can be converted into classification rules for easier understanding.

- Why are decision tree classifiers popular?
- ⇒ Don't require domain knowledge or parameter setting, making them suitable for exploratory knowledge discovery.
 - ⇒ Can handle multidimensional data.
 - ⇒ Have good accuracy.

Algorithms:

- ⇒ Popular algorithms like ID3 and C4.5 are used for generating decision trees based on univariate splits.
- ⇒ ID3 starts with all training samples at the node and recursively selects attributes to partition the samples based on information entropy.
- ⇒ To apply some of the methods, which are based on inductive - learning approach, several key requirements satisfied:

Decision tree Algorithms:

General algorithms:

For predicting the class of the given dataset, the algorithm starts from the root node of the tree.

⇒ Algorithm compares the values of root attribute with second attribute and based on the comparison, follows the branch and jumps to the next node.

⇒ For the next node, the algorithm again compares the attribute value with the other sub-nodes and moves further.

⇒ It continues the process until it reaches the leaf node of the tree.

Algorithm:-

Step-1:-

- Begin the tree with root node says S, which contains the complete dataset.

Step-2:-

- Find the best attribute in the dataset using ASM.

Step-03:

o Divide the S into subsets that contains possible values for the best attribute.

Step-04:

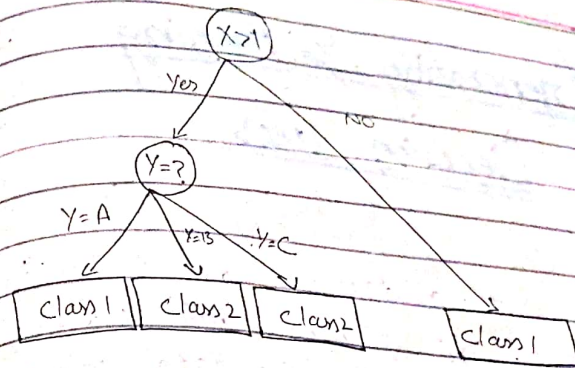
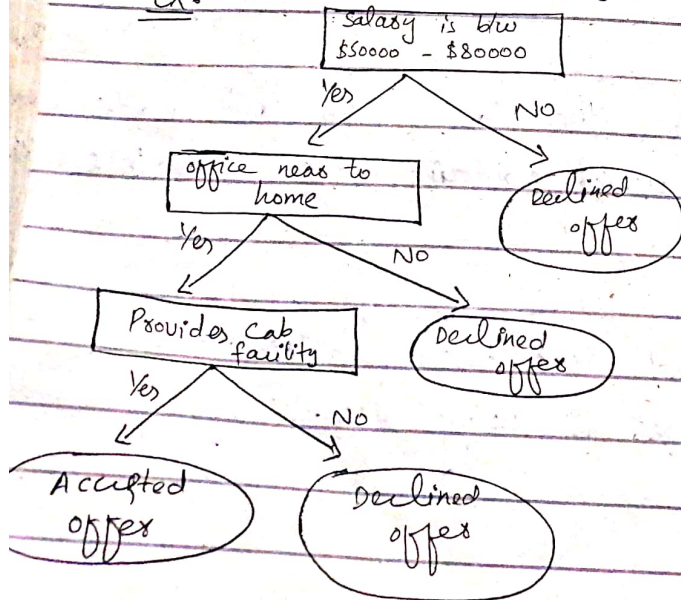
o Generate the decision tree node which contains the best attributes.

Step-04:

o Recursively make new decision trees using the subsets of the dataset created in subset step-3.

o Continue this process until a stage is reached where you cannot further classify the nodes and called the final node as a leaf node.

Ex:



Generating & Pruning

Decision Trees

Why decision trees are pruned?

⇒ When a decision tree is built, many of the branches will reflect anomalies in the training data due to noise or outliers.

⇒ Tree pruning methods address this problem of overfitting the data.

⇒ Such methods typically use statistical measures to remove the least-reliable branches.

⇒ Pruned trees tend to be smaller and less complex and, thus, easier to comprehend.

⇒ They are usually faster and better at correctly classifying independent test data than unpruned trees.

⇒ Main motive: minimize the overfitting of the models.

Definitions:

⇒ Discarding one or more subtrees and replacing them with leaves.
Simplifying a decision tree.

⇒ This increases the quality of a classification model.

⇒ Lowers predicted error rate.

How does tree pruning work?

⇒ There are two common approaches to tree pruning:

1. Prepruning
2. Postpruning

1. Prepruning

⇒ In prepruning approach, a tree is "pruned" by halting its construction early.

⇒ The stopping criteria is usually based on some statistical tests such as χ^2 test:

o if there is no significance differences in classification accuracy before and after division, then

represent a current node as a leaf node.

⇒ The decision is made in advance before splitting, and therefore this approach is called prepruning.

⇒ This technique is basically used before construction of decision tree.

⇒ When constructing a tree, measures such as statistical significance, information gain, Gini index, and so on, can be used to assess the goodness of a split.

⇒ If partitioning the tuples at a node would result in a split that falls below a preassigned threshold, then further partitioning of the given subset is halted.

⇒ However, difficulties in choosing the appropriate threshold.

⇒ High thresholds could result in oversimplified trees.

⇒ Whereas low thresholds could result in very little simplification.

Also known as deciding not to divide a set of samples any further under some conditions.

Post Pruning

⇒ The decision in this process of postpruning is made after the tree has been built.

⇒ It removes subtrees from a "fully grown" tree.

⇒ A subtree at a given node is pruned by removing its branches and replacing it with a leaf.

⇒ This follows the postpruning approach.

⇒ Also known as:

- Removing retrospectively some of the tree structure using selected arbitrary criteria.

Cost Complexity Pruning Algorithms

⇒ Cost Complexity pruning, which is used in CART, is a way to simplify decision trees after they have been built.

⇒ It looks at how complex the tree is and how many mistakes

it makes.

⇒ Starting from the bottom of the tree, it checks each part and calculates how complex it would be if that part was chopped off and replaced with a simplex one.

⇒ If making the tree simplex at that part would reduce its overall complexity, that part gets pruned.

Otherwise, it stays as it is.

⇒ This process helps to create a simplex yet effective decision tree.

Define pruning set?

⇒ In cost complexity pruning, a separate set of data, called a pruning set, is used to figure out how to simplify the tree.

⇒ The pruning set has labels for each example, just like the training set, but it is not the same data used to build the original tree or to check its accuracy.

Pessimistic pruning.

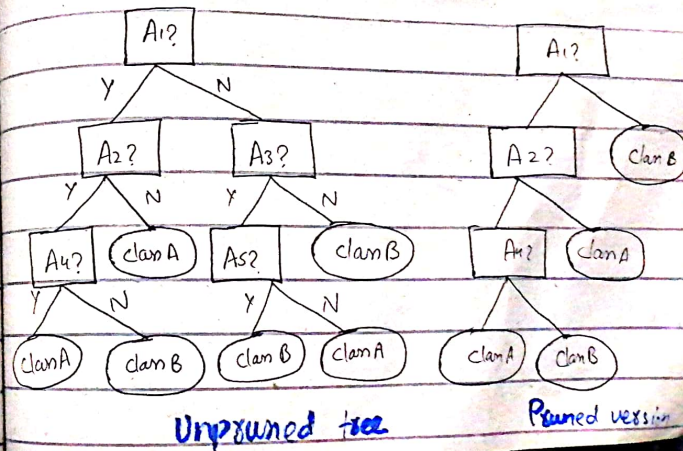
⇒ Pessimistic pruning as used in C4.5 is a way to simplify decision trees like cost complexity pruning but without needing a separate set of data for pruning.

⇒ Instead, it relies on the training data but adjusts for its overly positive estimate.

⇒ Because training data can make the tree seem better than it really is, pessimistic pruning adds a penalty to the error rates.

⇒ This ensures that the decision tree is not too complex or optimistic, leading to more reliable results.

Example:



Conclusions

- ⇒ Prepruning and postpruning may be interleaved for a combined approach.
- ⇒ Postpruning requires more computation than prepruning.
- ⇒ No single method has been found to be superior over all others.

CART Algorithm & Limit-Index

⇒ In 1984, a group of statisticians (L. Breiman, J. Friedman, R. Olshen and C. Stone) published the book Classification and Regression Trees (CART), which described the generation of binary decision trees.

⇒ CART adopt greedy approach in which decision trees are constructed in a top-down recursive divide-and-conquer manner.

⇒ it follow non-backtracking approach. It use Gini as ASM.

⇒ It is similar to CHS but it differs in tree structure, splitting criteria, pruning method, and handling of missing values.

Binary Splits

⇒ CART makes trees with only two-way splits, making things simpler.

⇒ it is good for splitting categories into two groups, but it can make

- Calculates the amount of probability of a specific feature that is classified incorrectly when selected randomly.
- trees harder to understand.

Gini index - Gini impurity.

⇒ CART uses the Gini diversity index instead of information-based criteria like CHS.

⇒ Gini-index decide how to split data.

⇒ it is a metric for the classification tasks in CART.

⇒ it works on categorical variables provides outcomes either "successful" or "failure" and hence conducts binary splitting only.

⇒ The degree of gini-index varies from 0 to 1, indicates random distribution - 0.5 → even distribution

⇒ it measures how pure or mixed the data is.

⇒ Mathematically we can write

Gini impurity as follows:

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

Where p_i is the probability of an object being classified to a particular class.

Reduction in impurity: $\Delta gini(A) = gini(D) - gini_A(D)$

• The attribute that maximizes the reduction in impurity is selected as the splitting attribute.

Splitting Criterion Calculations

if a data set D is split on A into two subsets D_1 and D_2 , the gini index.

⇒ The quality of a split is determined by the weighted sum of Gini indices of resulting subsets.

$gini_A(D)$ is defined as: $gini_A(D) = \frac{|D_1|}{|D|} gini(D_1) + \frac{|D_2|}{|D|} gini(D_2)$

Pruning Techniques

⇒ After the tree is built, CART uses pruning to prevent overfitting.

⇒ CART uses minimal cost complexity pruning, which is like trimming a tree to keep it simple but still useful.

Handling missing values

⇒ Unlike CHS, CART does not get bothered by missing info when splitting data.

⇒ it finds other clues to keep making good decisions.

Regression trees

⇒ CART can also make trees that predict numbers instead of categories.

⇒ it uses mean squared error to decide how good these trees are.

Robustness and invariance

⇒ CART is robust to outliers and noisy data, and its tree structure remains invariant to monotone transformations of independent variables.

Skew Sensitivity

⇒ C4.5 and CART's splitting criteria, information gain and gini index, are considered skew-sensitive.

⇒ To fix this, a new method called Hellinger distance might work better.

Steps of Algorithm:

Step-01

o Calculate the gini index of the dataset.

$$\text{Gini-index}(T) = 1 - \sum_{i=1}^m P_i^2$$

Step-02

o Compute Gini-index for each attribute and each of the subset in the attribute.

Step-03

o Choose the best splitting subset which has minimum gini-index for an attribute.

Step-04

o Compute Gini or best splitting subset of that attribute.

⇒ Repeat the same process for the remaining attributes in the dataset such as for interactivity, practical knowledge and communication skills.

Examples

= Data about customers including age, income, and whether they bought a product.

Customer	Age (years)	Income (USD)	Buys products?
1	25	30,000	
2	35	40,000	Yes
3	45	80,000	Yes
4	20	20,000	Yes
5	55	60,000	No
6	50	100,000	Yes
7	60	90,000	No
8	30	50,000	Yes
			No

1. Splitting Criteria

o CART chooses features and split points to divide the data.

Calculates the $Gini$ for each possible split and selects one that minimizes impurity.

2. Building the tree
→ CART chooses the age feature and splits the data into two subsets based on $age \leq 40$ and $age > 40$.

3. Recursive splitting

→ CART repeats the splitting process for each subset, creating a tree structure.

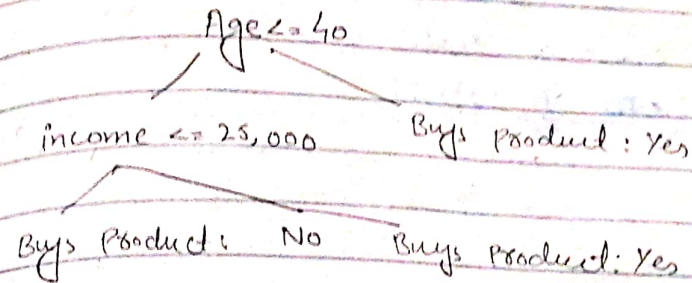
→ For the $age \leq 40$ subset, CART may further split based on income.

→ For example, it may split the data into $income \leq 25,000$ and $income > 25,000$.

4. Stopping criteria

→ CART stops splitting when a predefined stopping criterion is met, such as reaching a maximum tree depth or minimum number of samples in a node.

5. Pruning
→ After the tree is built, CART prunes it to prevent overfitting.
→ It removes parts of the tree that do not significantly improve predictive accuracy.



Interpretation:-

→ Customers aged 40 or younger with income $\leq 25,000$ are unlikely to buy the product.

→ Customers aged 40 or younger with income $> 25,000$ are likely to buy the product.

Pros of CART:

- Robustness to outliers & noisy data.
- Easy to understand decision tree.
- Classification and regression trees
- are non-parametric and non-linear.
- Implicitly perform feature selection.
- outliers have no meaningful effect on CART.

Cons of CART:

- Overfitting
- High variance
- Low bias
- The tree structure may be unstable.
- Parametric & skew-sensitivity.

Applications:-

- ⇒ For quick data insights.
- ⇒ In Blood Donors Classification.
- ⇒ ~~for~~ for ecological data.

Advantages of Decision Trees and Decision rules:

- ⇒ Decision tree and decision-tree based models are known for their simplicity, readability, and quick generation.
- ⇒ Do not rely on attribute value distributions or attribute independence.

Limitations:

- Split data into regions, but can be tricky.
- Regions get complicated.
- More complex models need more regions. Need lots of training data for good results.
- Dealing with tricky patterns:
- ⇒ Sometimes patterns are not straightforward.
- This makes models more complicated and prone to mistakes.
- ⇒ Dealing with non-linear classification problems.
- ⇒ Introduction of new attributes for simplification.

⇒ Limited representation of non-orthogonal data.

⇒ High error rates with increased complexity.

⇒ Risk of overfitting

⇒ Limited representations of classifiers.
